REVERSE ENGINEERING · APPLE POWER ACCESSORIES

Apple USB-C and MagSafe 3 power path

Three processors, three firmware objects, one name

The cable's three releases extracted, the container format recovered from Apple's own parser, and the cross-version facts established without touching a device.

REPORT	VERSION	DATE	CLASSIFICATION
RR-2026-1114	1.1	2026-11-11	Internal
TARGET	BASELINE	PREPARED BY	TEMPLATE
3.2.0	3.1.8	Sein	v1.2.0

DOCUMENT CONTROL

REPORT	Apple USB-C and MagSafe 3 power path
REPORT ID	RR-2026-1114
VERSION	1.1
DATE	2026-11-11
CLASSIFICATION	Internal
BASELINE	MagSafe 3 · 3.1.8
TARGET	MagSafe 3 · 3.2.0
PREPARED BY	Sein
SOURCE MATERIAL	Apple public UARP assets, hpmdiagnose register dumps, IORegistry captures
RENDERER	Typst · Research Reports template 1.2.0

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1 Executive summary

Three firmware objects share the phrase “the charger firmware” and nothing else. This report separates them on a live Apple Silicon MacBook Pro, extracts the MagSafe 3 cable’s three public releases, recovers the container format from Apple’s own parser, and establishes what did and did not change between them — read-only throughout.

VERDICT

The host power path, the container format and the cross-version byte facts are established. No host-observable behaviour change is demonstrated, and semantic equivalence between 3.1.12 and 3.2.0 stays `OPEN`.

KEY FINDINGS

01 **Three objects, one name** — HPM controller firmware `FW003.115.00`, cable 3.2.0 (PID `0x7800`, bcdDevice `0x3200`) and adapter `1030030` are distinct. Only the first two are host-verifiable; the adapter value is reported, not read.

02 **Container recovered, not guessed** — The UARP SuperBinary layout came out of Apple’s own parser in `uarpd`: a 44-byte header, 40-byte payload records, and six range checks that all four acquired files pass.

03 **One literal changed** — `0x05AC0002` disappears and `0x05AC0000` gains an occurrence in 3.2.0. The power-negotiation table is byte-identical in all six copies across all three builds.

04 **The dispatcher idiom moved** — A 14-case `mov pc` table is present in both 3.1.x builds and absent from 3.2.0, which shows 10 detected sites of the same mechanism with different tables.

3	4	38.8 %
CABLE RELEASES EXTRACTED	HPM CONTROLLERS MAPPED	OF IMAGE INSIDE ANALYSED FUNCTIONS

2 The three objects

Calling all of this “charger firmware” is what makes the topic confusing. They are three different processors with three different update paths.

TABLE 1

Firmware objects in the power path

OBJECT	IDENTIFIER	HOW IT WAS READ
HPM controller	`FW003.115.00`	From the kext’s own strings and register file
MagSafe 3 cable	3.2.0 / PID `0x7800` / `0x3200`	Register `0x49` and the in-file version
70 W adapter	`1030030` / ID `0x701c`	Host-reported only: no adapter firmware obtained

Source: IORegistry and `IOAccessoryManager` captures in `evidence/`, `hpmdiagnose` register dumps, and `AppleHPM` decompilation.

The distinction matters for any security claim. The host can read the cable’s version out of the negotiation it already performs; it cannot read the adapter’s flash, and nothing in the acquired data shows otherwise.

3 The host power path

Four HPM instances exist: three USB-C ports at RID 0, 1, 2 and the MagSafe port at RID 5. The charger is on RID 5, and the winning power option is 20 V at 3.390 A, which is 67.80 W.

ACCESS

The full HPM register file is readable without root by `hpmdiagnose`; access is gated by the `com.apple.USBCEntitlement` entitlement, not by privilege. That is the practical ceiling of noninvasive inspection, and it is higher than the usual “needs root” assumption.

What the host exposes is a contract: a set of fixed power options, one of which is selected. That is a negotiation result, not a firmware property, and the two should never be tabulated as the same kind of fact.

4 Acquiring the firmware

Apple publishes accessory firmware under `com.apple.MobileAsset.UARP.<model>` on `updates.cdn-apple.com`. Each download was verified against the SHA-1 Apple itself publishes in `_Measurement`, before anything was unpacked.

TABLE 2

Acquired cable releases

ASSET	SHA-1 (TRUNCATED)
A2363_magsafe3_3.1.8_10M569.zip	3722c634e671…
A2363_magsafe3_3.1.12_10M1543.zip	a10fd2efd5cc…
A2363_magsafe3_3.2.0_10M13722.zip	4db70dec4835…
A2452_140W_1.4.90_10M13554.zip	60b42b01ce02…

Source: `firmware/MANIFEST.txt`; SHA-1 truncated here, full values and URLs in the manifest.

The 70 W adapter is the exception, and the negative is scoped rather than absolute: three enumeration methods failed — `mesu` path probes, a Pallas lookup including a deliberately fake control type, and a broad type sweep. That is evidence about Apple's public catalogue, not proof the firmware does not exist.

5 Container format

The layout was recovered from Apple's parser (`expandSuperBinaryHeader` and the payload-header walker in `uarpd`), not inferred from hex. That distinction is why the extraction is trustworthy: the structure is the one the host enforces.

A 44-byte header carries the payload extent; each payload is described by a 40-byte record with 4CC-tagged TLV metadata. On all four acquired files, `dataOffset + dataLength` equals the declared extent, and six range checks pass.

METHOD

An earlier attempt walked the file as generic TLVs and produced a plausible, wrong structure. Recovering the parser changed the answer. Guessing a container format from byte patterns is the failure mode this step avoids.

6 Image anatomy

Each payload opens with a 384-byte `C99` wrapper, then four words: stack pointer `0x20002000` and three pointers. In 3.2.0 the word at offset `0x04` is `0x0001111d`, whose

Thumb target `0x1111c` holds `0x4005xxxx` peripheral addresses, so the target is constant data, not a reset handler.

The load base is `OPEN`. The 140 W control image has a similar opening, but its pointers `0xec89` and `0xec71` exceed its own `0xea00` payload, which rules out a base-zero mapping for that image without ruling out other bases — a base of `0x400` would place both inside. It is not corroboration, and it is not used as any.

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7 What changed, and what the numbers can say

Three measurements survive scrutiny, and each is deliberately narrow.

TABLE 3

Cross-version observations

OBSERVATION	3.1.8	3.1.12	3.2.0
Fixed PDO table	5/9/15/20 V @ 3 A	same	same
`0x05AC0002` occurrences	1	1	0
`0x05AC0000` occurrences	6	6	7
`mov pc` dispatch sites	14	14	10
14-case table	present	present	absent
Bytes inside functions	38.8 %	39.4 %	43.3 %

Source: `scripts/diff_verify.py`, `scripts/dispatcher_census.py`, `scripts/coverage.py`; all re-derivable.

Aligned 16-byte window overlap is 73.1 % between the 3.1.x pair and 5.9 % between 3.1.12 and 3.2.0. That measures byte layout, not logic: reordered functions, moved literal pools and different register allocation all lower it with unchanged source. Reading it as “a rewrite” was an error, and it is withdrawn.

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8 The three 2026 claims, examined

- Hard-reset handling changed — `REFUTED`. The switch that could carry such a claim covers message types 9 to 22, and Hard_Reset has no MessageType: it is a vSafe0V protocol action. A message switch cannot speak to it.
- Soft-reset escalation — `OPEN`. No ladder was found in any of the three images. Absence in the analysed 39 % is not absence in the firmware.
- Power negotiation changed — not supported. The PDO table is identical in all six copies; only its address moves, so a fixed-offset read returns garbage.

9 The host stack

`uarpd` carries the message format: a six-byte header of three big-endian `u16` fields, a vendor tag `09 BC 63` followed by a type and value, and a 39-entry opcode table. The updater's Mode 7 address-space read exists in the binary, but reaching it means driving the accessory into update mode — an authorised experiment, not a routine step. Whether that verb can touch flash is `OPEN`.

10 Security posture

The trust boundary that can be stated today is narrow: the host verifies the manifest hash before applying anything, and the accessory image carries no signature this analysis could identify. Whether a signature is enforced below that hash, whether anti-rollback exists, and what the three 32-byte `C99` header fields are, all remain `OPEN`. Nothing here is reported as a vulnerability.

11 Limits

Coverage is 38.8 / 39.4 / 43.3 % of the images, and the paths that matter — the control-message switch and the Apple SVID literal sites — sit outside the analysed set. Per-function matching is not reported at all: both calibrated runs were void, and their causes are recorded. The adapter firmware was never obtained, so adapter behaviour is host-observed only.

12 Method

Evidence came from IORegistry and `IOAccessoryManager` captures, a deterministic `hpmdiagnose` dump, Apple's public manifests, and static analysis. One label scale throughout: `OBSERVED`, `REPRODUCED`, `INFERRED`, `UNVERIFIED`, `REFUTED`, `OPEN`.

SCOPE

All hardware interaction was read-only: no flashing, no update trigger, no accessory command. Proprietary images are not redistributed; hashes and acquisition notes are.

13 Appendix A — claim register

Every claim is checked across the research documents by `scripts/verify_claims.py`, which fails on a missing statement, a conflicting label in the same paragraph, a withdrawn phrase in a conclusion, or a broken evidence reference.

- **C01** [OBSERVED] The charger negotiates 20 V at 3.390 A (67.80 W) on the MagSafe port.
- **C02** [OBSERVED] Four HPM controllers are present at RID 0, 1, 2 and 5; RID 5 carries the MagSafe port.
- **C03** [OBSERVED] HPM controller firmware FW003.115.00, the MagSafe cable at 3.2.0 and the adapter value 1030030 are three distinct objects, not revisions of one image.
- **C04** [REPRODUCED] The UARP SuperBinary container is a 44-byte header with 40-byte payload records; dataOffset plus dataLength reconciles in all four files.
- **C05** [OBSERVED] Three MagSafe 3 releases were acquired and each SHA-1 matches Apple's manifest _Measurement.
- **C06** [REPRODUCED] The power-negotiation table content (5/9/15/20 V at 3.00 A) is identical in all builds while its address moves.
- **C07** [OBSERVED] Exactly one Apple SVID literal changes across the three releases: 0x05AC0002 disappears and 0x05AC0000 gains an occurrence.
- **C08** [REPRODUCED] Between 3.1.12 and 3.2.0 the compared bodies share 11 non-trivial literals, lose 7 and gain 1; 28 is present only in 3.2.0.
- **C09** [REFUTED] The vector word at 0x0001111d points to 0x1111c, which is constant data and not a reset handler.
- **C10** [OBSERVED] The 140 W control image's pointers 0xec89 and 0xec71 exceed its 0xea00 payload, which rules out a base-zero mapping only; a non-zero base such as 0x400 would place them inside.
- **C11** [REPRODUCED] The frame header formula (sop<<4)|ceil(bytes/4) reproduces the observed 0x21 for SOP selector 2 and a 4-byte payload; the meaning of the sop field outside the observed selector set is open.
- **C12** [OBSERVED] A 14-case mov-pc dispatch table with 6 unique targets and 8 of 14 cases to one target is present in both 3.1.x builds and is not detected in 3.2.0.
- **C12b** [OPEN] The 15-case and 13-case tables in 3.2.0 are candidates only; no correspondence to the 3.1.x switch is shown.
- **C13** [REFUTED] Claims that 3.2.0 replaced or reorganised the dispatcher, or is a rebuild rather than a patch, are withdrawn.
- **C14** [UNVERIFIED] The Apple SVID literal sites are not established as a UVDM builder; that identification remains unverified.
- **C15** [REFUTED] The 5.9 % / 73.1 % aligned-window overlap figures measure layout; the inference from them to a rewrite is refuted.
- **C16** [REPRODUCED] Ghidra function coverage is 38.8 / 39.4 / 43.3 % of image bytes, and the switch and literal sites lie outside the analysed set.
- **C17** [REFUTED] Hard-reset handling change is refuted: Hard_Reset has no MessageType, so a 9-22 message-type switch cannot speak to it.
- **C18** [OPEN] No soft-to-hard reset escalation ladder was found; the question stays open across all three images.

- **C19** [OBSERVED] 70 W adapter firmware was not found by three enumeration methods (mesu path probes, Pallas lookup with a control type, and a broad type sweep); this is a scoped negative, not proof of nonexistence.
- **C20** [OPEN] Signature enforcement below the manifest hash, anti-rollback, and whether the Mode 7 read verb can touch flash are all open.
- **C21** [OBSERVED] hpmdiagnose reads the full HPM register file unprivileged; access is entitlement-gated, not root-gated.
- **C22** [OPEN] Semantic equivalence of 3.2.0 to 3.1.12 is undeterminable at the achieved coverage and is open.
- **C23** [OPEN] The entry-point question is OPEN: offset 0x10 is the empirical start, not a proven entry, and 0x1111c holds peripheral addresses.
- **C24** [REFUTED] The 140 W control image does not corroborate a base-zero layout for the MagSafe images.
- **C25** [OBSERVED] The three firmware objects are not equally established: the adapter value is host-reported only, and the cable version is a single in-file source consistent with register 0x49.
- **C26** [REPRODUCED] 3.2.0 contains 10 detected mov-pc dispatch sites, not 15.
- **C28** [REPRODUCED] The C99 inner header is a two-record SHA-256/CRC-32 hash chain over the payload body, record 3, record 4, plus a header self-hash; all five relations reproduce byte-exactly on all three payloads.
- **C29** [REPRODUCED] The flat base0 images equal payload[0x180:] — the 384-byte C99 header is stripped in extraction, not absent from the asset.
- **C30** [REPRODUCED] SHA-256 (FIPS 180-4 K table + IV, twice per image) and a complete secp256r1 parameter block (p, b, Gx, Gy, n) are present in all three images.
- **C31** [INFERRED] The 3.1.x dispatcher's second-stage 17-case switch has a verified 3.2.0 successor (0xad2a/0xad7e/0xab64) whose only structural change is the index width (ldrb -> ldrh).
- **C32** [OPEN] The 3.1.x dispatcher island has zero static inbound edges and even mov pc table targets (ARM-state request on v7-M); it is very likely a stale or dead copy, and the live dispatcher's location is open.
- **C33** [OBSERVED] Of the 155 non-stub calibrated 1:1 pairs 3.1.12->3.2.0, 124 are identical modulo relink and 31 are substantive, including a signed->unsigned comparison change (0xa018->0xa020).
- **C34** [OBSERVED] The 3.1.x dispatcher default arm (including Soft_Reset, case 13) rebuilds the source-cap buffer and sets a port state byte; no Accept transmit or CC/PHY write is decoded in that body.
- **C35** [OBSERVED] The FSM state-name string spm_host_src_hardreset exists in 3.1.8 and 3.1.12 and is absent from 3.2.0, whose pool retains four reordered states.
- **C36** [OPEN] No version-threshold constant or cmp.w instruction exists in any image; at 46-56% function coverage this negative is bounded, not conclusive — anti-rollback is unresolved.
- **C37** [OBSERVED] VDMs is one 4CC command tag among many; UARP uses CFUa/CFUp/CFUs as runtime 4CC arguments. The cable images contain an SVDM dispatcher (0xb34c) with no 4CC compare; CFU consumption below the app image is open.